CHAPTER

Feed and Process Pumps

Feed, diluate and concentrate pumps on each SonixED skid

IN THIS CHAPTER

9.1 Overview and function

Each SonixED™ skid carries three process pumps — feed, diluate and concentrate — that circulate water through the stack(s) at the required flow and pressure. Pumps are magnetic-driven (sealless) and inverter-(VFD-)controlled so flow can be adjusted without throttling valves.

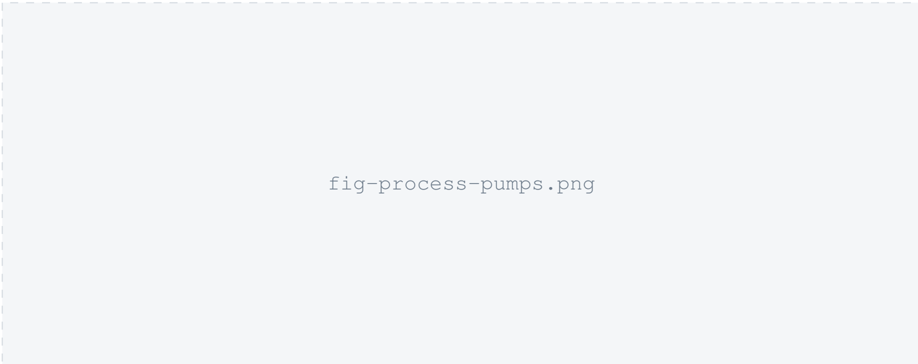

Figure 9.1. Suggested figure: photograph of the three process pumps on a skid, labeled feed/diluate/concentrate, with their VFD units visible.

9.2 Key specifications

- Always 3 pumps per skid (feed, diluate, concentrate).
- Magnetic-driven, inverter-driven, polypropylene/stainless-steel construction, seawater-compatible.
- Pretreatment feed pump (pilot): Grundfos CRN 10-5 A-FGJ-A-E-HQQE, flanged connections.
- Rated hydraulic flow per stack: 24 m^3/day (SonixED-S datasheet).
- Each pump is driven by its own VFD (control tags `VFD_1_D1...VFD_6_E2` in the pilot configuration).

9.3 Operation

Pump speed is set by the PLC via the VFDs to hold the process at its target flow/pressure; TODO no document confirms the exact control loop (open-loop speed vs. closed-loop flow/pressure control). Manual speed override, if available, is reached via the `Manual` mode described in chapter 6.

9.4 Maintenance

TODO: no pump-specific maintenance schedule (bearing/seal inspection, impeller check interval) was found. Magnetic-driven pumps have no shaft seal to replace, which reduces routine maintenance versus a conventional centrifugal pump, but confirm this against the actual pump model fitted.

9.5 Troubleshooting

Each pump's VFD reports a `FaultWord` that should be checked first on any pump-related alarm. Typical causes to check, in order:

- **Pump not running / VFD fault:** check `FaultWord` on the relevant VFD tag; check incoming supply and control-voltage presence (chapter 14).
- **Low flow / high pressure:** check for blockage upstream (chapter 8) or downstream (stack fouling, chapter 10) before assuming a pump fault.
- **Pump running but no output:** check for air lock (magnetic- driven pumps are not self-priming) — TODO confirm priming procedure.